

live_monitor — Full Execution Flow

Live polling cycle + ML inference + Postgres persistence + Retrain pipeline

Target storage: Postgres via Backend APIs
SQLite = legacy (retrain cutover remaining)

Ingest / Buffer ML Inference Evaluation Postgres / Backend Retrain / Training Model Artifacts (.pkl) Local debug :8001

A) LIVE CYCLE (every ~10 seconds) — main.py → run_cycle()



live_monitor — Decision Points & Sequence Detail

Page 2 — what happens when steps succeed / skip / fail

Live cycle decisions (run_cycle)



Retrain sequence (manual) — scripts in order



End-to-end story in one sentence

Every 10s the live cycle turns sensor polls into features, ML state, anomaly, and baseline evaluation, then stores everything in Postgres for the Frontend Operations Center.
Periodically (or manually), retrain rebuilds models from historical raw windows and writes new .pkl files, which the next live_monitor restart uses for better state/anomaly detection.
Postgres-only cutover = make the retrain history source the same Postgres the live cycle already writes to.